

BARRY MENGLONG YAO

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Education

UC Davis

Ph.D. in Computer Science

- Advisor: Dr. Lifu Huang

Aug. 2022 – Mar. 2027

Best Paper Honorable Mention at SIGIR 2023

University at Buffalo, The State University of New York

Master of Science in Computer Science and Engineering

- Advisor: Dr. Changyou Chen

Aug. 2020 – Jun. 2022

CSE Best MS Research Award

Jilin University

Bachelor of Science in Computer Science and Technology

Sept. 2012 – Jun. 2016

The Third Prize Scholarship

Research Summary

My research interests are in Multimodal Learning and NLP, with a focus on **Data-Efficient Post-Training of Multimodal Foundation Models**. I have publications in top-tier venues such as ACL, EMNLP, EACL, SIGIR, and AAAI, with ~300 citations, including a **Best Paper Honorable Mention** at SIGIR 2023. My representative works span *Error-Driven Data Selection for Large Multimodal Model Post-Training*, *Data-Centric Learning Dynamics Analysis for Continual Pre-Training of Large Language Models*, and *Adaptive Data Synthesis for Long-Horizon Multimodal Planning*.

Experience

[1] Applied Scientist Intern, Oracle

advised by Dr. Mihaela Bornea, Dr. Avi Sil

- Self-Evolving Disambiguation over Knowledge Base

June. 2026 – Sept. 2026

[2] Applied Scientist Intern, Amazon Alexa AI

advised by Dr. Xiyao Ma, Dr. Zhongkai Sun, Dr. Jie Hao, Dr. Jun Yin, Dr. Lichao Wang

- LLM Agent Chain-of-Thought Optimization for Multi-Turn Tool Calling (SFT + RL)

June. 2025 – Sept. 2025

[3] Applied Scientist Intern, Amazon Alexa AI

advised by Dr. Cynthia Lu, Dr. Ziming Li, Dr. Jinseok Nam, Dr. Derek Liu

- LLM Inference Latency Optimization via Adaptive Sparsity

May. 2024 – Aug. 2024

[4] Software Engineer, WeBank Co., Ltd

- Patent: An exception handling framework: State Machine framework. Youzhi Liao, Barry Menglong Yao and Min Jiang.

Aug. 2016 - July 2020

Publications & Manuscripts

[1] How Do Large Language Models Learn Concepts During Continual Pre-Training?

- Barry Menglong Yao, Sha Li, Yunzhi Yao, Minqian Liu, Zaishuo Xia, Qifan Wang, Lifu Huang.
- EMNLP 2026.

[2] Error-driven Data-efficient Large Multimodal Model Tuning

- Barry Menglong Yao, Qifan Wang, Lifu Huang.
- ACL 2025.

[3] Self-Evolving Learning for Long-Horizon Multimodal Planning

- Barry Menglong Yao, Zaishuo Xia, Dechen Gao, Yubei Chen, Qifan Wang, Lifu Huang.

[4] End-to-End Multimodal Fact-Checking and Explanation Generation: A Challenging Dataset and Models

- Barry Menglong Yao, Aditya Shah, Lichao Sun, Jin-Hee Cho and Lifu Huang.

- **Best Paper Honorable Mention** at SIGIR 2023.
- [5] **Ameli: Enhancing Multimodal Entity Linking with Fine-Grained Attributes**
- Barry Menglong Yao, Sijia Wang, Yu Chen, Qifan Wang, Minqian Liu, Zhiyang Xu, Licheng Yu, Lifu Huang.
 - EACL 2024 (Oral).
- [6] **FedIOD: A Federated Learning Framework with Input-Output Ensemble Distillation**
- Xuan Gong, Shanglin Li, Yuxiang Bao, Barry Menglong Yao, Yawen Huang, Ziyang Wu, Baochang Zhang, Yefeng Zheng, David Doermann.
 - AAAI 2024.
- [7] **A survey on mechanistic interpretability for multi-modal foundation models**
- Zihao Lin, Samyadeep Basu, Mohammad Beigi, Varun Manjunatha, Ryan A Rossi, Zichao Wang, Yufan Zhou, Sriram Balasubramanian, Arman Zarei, Keivan Rezaei, Ying Shen, Barry Menglong Yao, Lifu Huang.
- [8] **Scientific Hypothesis Generation and Validation: Methods, Datasets, and Future Directions**
- Adithya Kulkarni, Fatimah Alotaibi, Xinyue Zeng, Longfeng Wu, Tong Zeng, Barry Menglong Yao, Minqian Liu, Shuaicheng Zhang, Lifu Huang, Dawei Zhou.
- [9] **KnowledgeBot: Improving Assistive Robot for Task Completion and Live Interaction via Neuro-Symbolic Reasoning**
- Minqian Liu, Ying Shen, Barry Menglong Yao, Sijia Wang, Jingyuan Qi, Zhiyang Xu, Lifu Huang.
 - In Alexa Prize Simbot Challenge Proceedings, 2023. (Ranked 4/10 in Alexa Prize SimBot Challenge.)
- [10] **HOKIEBOT: Towards personalized open-domain chatbot with long-term dialogue management and customizable automatic evaluation**
- Ying Shen, Jingyuan Qi, Sijia Wang, Barry Menglong Yao, Minqian Liu, Zhiyang Xu, Trevor Ashby, Lifu Huang.
 - In Alexa Prize SocialBot Grand Challenge 5 Proceedings, 2023.
- [11] **An exception handling framework: State Machine framework. (Patent)**
- Youzhi Liao, Barry Menglong Yao and Min Jiang.

Honors & Awards

Best Paper Honorable Mention Award, SIGIR 2023
CSE Best MS Research Award, UB CSE Department
The Third Prize Scholarship, Jilin University 2013
The Third Prize Scholarship, Jilin University 2014